Version November 11, 2026
Preprint typeset using LaTeX style openjournal v. 09/06/15

IS ΛCDM COMPATIBLE WITH THE OBSERVED CMB TEMPERATURE CORRELATION FUNCTION?

Álvaro Méndez Rodríguez de Tembleque
Università degli Studi di Padova, Department of Physics and Astronomy "Galileo Galilei"

Yago Ascasibar
Universidad Autónoma de Madrid, Departamento de Física Teórica and
Centro de Investigación Avanzada en Física Fundamental (CIAFF-UAM), 28049 Madrid, Spain
Version November 11, 2026

ABSTRACT

We investigate the statistical properties of the cosmic microwave background (CMB) angular correlation function $\xi(\theta)$ using Planck legacy data. Three complementary statistics are studied: the antipodal correlation ξ_{180}, the interval-averaged correlation $\langle\xi\rangle_{\theta_a}^{\theta_b}$ between θ_a and θ_b, and the mean squared correlation $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$. These statistics are combined using the Aggregated Cauchy Association Test (ACAT) to quantify their overall significance, mitigating the 'look-elsewhere' effect. Our analysis shows that the observed CMB temperature is systematically less correlated than the predictions of the ΛCDM model at all—not only large—angular scales. On the other hand, the statistical significance of the deviation is mild (one-sided p-value $p = ...$, equivalent to $\sim ...\sigma$), and therefore it can hardly be qualified as a cosmological tension.

Subject headings: cosmic microwave background, angular correlation function, large-angle anomalies, Planck, ΛCDM ... CHOOSE FROM LIST

1. INTRODUCTION

Since the first measurements of the CMB temperature anisotropies, the angular two-point correlation function $\xi(\theta)$ has attracted particular attention. At variance with the amplitudes C_l of the power spectrum, this statistical descriptor highlights the largest angular scales, thus probing the longest wavelength perturbations accessible within the observable universe.

Early analyses of the COBE-DMR data already suggested that $\xi(\theta)$ was unexpectedly close to zero for separations larger than approximately 60° (Hinshaw et al. 1996). Although the statistical power of COBE was limited, this feature was subsequently confirmed with much higher precision by the first-year WMAP observations (Spergel et al. 2003), and the descriptor

$$S_{1/2} = \int_{-1}^{1/2} \xi^2(\theta)\, \mathrm{d}(\cos\theta), \tag{1}$$

was introduced to quantify the magnitude of the effect, measuring the integrated squared correlation function over angular separations larger than 60° (Spergel et al. 2003). The observed value of $S_{1/2}$ was found to lie in the extreme tail of the ΛCDM distribution generated from Monte Carlo realisations, establishing the lack of large-angle correlations as one of the most persistent large-scale anomalies in the CMB.

A series of subsequent studies strengthened this conclusion based on successive WMAP data releases. In particular, analyses of masked sky maps pointed out that the observed correlation function is nearly zero for $\theta \gtrsim 60°$, with probabilities typically below 10^{-3} under the standard cosmological model (Copi et al. 2006, 2009, 2013). These works emphasized that the discrepancy cannot be explained simply by the low quadrupole amplitude alone, but instead reflects an unusual covariance structure among the lowest multipoles. They also noted connections with other large-scale anomalies, including the quadrupole-octopole alignment and hemispherical asymmetries.

On the contrary, Efstathiou et al. (2010) argued that estimators based directly on masked-sky correlation functions can exaggerate the discrepancy with ΛCDM, and showed that full-sky maximum-likelihood reconstructions yield results more consistent with theoretical expectations. Extending this argument, Pontzen & Peiris (2010) considered a broader class of anisotropic Gaussian models and concluded that the observed correlation function does not constitute compelling evidence for new physics once the full space of possible covariance structures is taken into account. These studies highlighted the importance of estimator choice, a posteriori statistic selection, and the definition of the theoretical ensemble used when assessing significance.

The anomaly has also motivated the development of alternative statistical descriptors of large-scale correlations. Direction-dependent generalizations of the correlation function were introduced to investigate whether the missing power is localized on the sky rather than uniformly distributed. Zhang (2012) constructed directional versions of both $\xi(\theta)$ and $S_{1/2}$, showing that a large fraction of the full-sky signal originates from regions associated with the quadrupole and octopole extrema. Other approaches include bipolar harmonic decompositions (Joshi et al. 2009), generalized isotropic angular correlation functions (Dipanshu et al. 2023), geometric representations of the two-point function (Froes et al.

alvaro.mendezrodriguezdetembleque@studenti.unipd.it
yago.ascasibar@uam.es

2015), and all-angle statistics such as

$$S_1 = \int_{-1}^{1} \xi^2(\theta)\, \mathrm{d}(\cos\theta), \tag{2}$$

proposed to reduce the dependence on the a posteriori choice of the 60° threshold (Gruppuso 2014).

Regarding the connection to other low-multipole features, Sarkar et al. (2010) found essentially no correlation between the suppression of $\xi(\theta)$ and the well-known quadrupole-octopole alignments within ΛCDM realisations. The simultaneous occurrence of both anomalies is therefore substantially rarer than each of them considered individually. Similarly, Kim & Naselsky (2011) argued that the observed odd-parity preference at low multipoles provides a phenomenological explanation for the deficit of large-angle correlations, although no convincing physical origin was identified. Analyses of Planck polarization maps have revealed a similar suppression of large-angle correlations, with significances comparable to those found in temperature data (Chiocchetta et al. 2020). If confirmed, such results would suggest that the deviation from the predictions of the ΛCDM model is not solely a consequence of temperature-specific systematics and may instead reflect a more fundamental property of the largest observable scales.

Despite nearly two decades of study, there is still no consensus regarding neither the statistical significance nor the physical interpretation of the lack of temperature correlation. In this work we revisit this issue based on Planck 2018 data. In addition to the $S_{1/2}$ and S_1 statistics defined above, we also consider the antipodal correlation ξ_{180}, the average correlation $\langle\xi\rangle_{\theta_a}^{\theta_b}$ over 4 contiguous angular intervals, and the mean squared correlation $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ over the same intervals. We construct random Monte Carlo realisations of the best-fit ΛCDM temperature spectrum to account for cosmic variance, and we combine the individual p-values of all statistics using the Aggregated Cauchy Association Test (ACAT) to obtain a global measure of the tension between the theoretical prediction and the observational measurements.

... Section...

2. METHODOLOGY

2.1. *Statistical descriptors*

It is customary to express the temperature anisotropies

$$\frac{\Delta T(\hat{n})}{T_0} = \sum_{\ell,m} a_{\ell m} Y_{\ell m}(\hat{n}) \tag{3}$$

in terms of spherical harmonics $Y_{\ell m}(\hat{n})$ with real coefficients $a_{\ell m}$. For a statistically isotropic Gaussian random field, the two-point statistics are fully encoded in the amplitudes of the power spectrum $C_\ell = \langle |a_{\ell m}|^2 \rangle$ or, equivalently, the scaled spectrum $D_\ell = \ell(\ell+1)C_\ell/(2\pi)$, which is approximately flat for a scale-invariant spectrum of primordial fluctuations. The angular two-point correlation function may then be obtained by transforming back to real space, yielding the polynomial expansion

$$\xi(\theta) = \sum_{\ell} \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos\theta), \tag{4}$$

where P_ℓ denote the Legendre polynomials.

TABLE 1: Angular intervals used for the statistics $\langle\xi\rangle_{\theta_a}^{\theta_b}$ and $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$, obtained from the zeroes of the observed (Planck) correlation function.

Angular range	$[\cos\theta_b,\ \cos\theta_a]$
$[0°,\ 30°]$	$[0.865,\ 1.000]$
$[30°,\ 92°]$	$[-0.039,\ 0.865]$
$[92°,\ 154°]$	$[-0.901,\ -0.039]$
$[154°,\ 180°]$	$[-1.000,\ -0.901]$

In addition to the descriptors $S_{1/2}$ and S_1 proposed in the literature, let us also consider the following statistical probes:

Antipodal correlation: $\xi_{180} \equiv \xi(\theta = 180°)$ measures the correlation between antipodal points on the sky. Because it is a pointwise evaluation of $\xi(\theta)$, it is relatively insensitive to masking effects at intermediate scales, and it provides a direct test of the alleged near-vanishing of large-angle correlations at the largest possible separation.

Interval-averaged correlation: $\langle\xi\rangle_{\theta_a}^{\theta_b}$ corresponds to the mean of $\xi(\theta)$ over the angular interval $[\theta_a, \theta_b]$

$$\langle\xi\rangle_{\theta_a}^{\theta_b} = \frac{1}{\mu_a - \mu_b} \int_{\mu_b}^{\mu_a} \xi(\theta)\, d\mu \tag{5}$$

where $\mu_\theta \equiv \cos\theta$ accounts for the surface area of the sky as a function of polar separation angle, as in $S_{1/2}$ and S_1. At variance with these traditional descriptors, ξ_{180} and $\langle\xi\rangle_{\theta_a}^{\theta_b}$ preserve the information about the sign, distinguishing between correlated and anti-correlated temperature fluctuations. Therefore, to partition the full range $0° \leq \theta \leq 180°$ into physically motivated regions, we select the null-crossings $\xi(\theta) = 0$ where the observed correlation function (Aghanim et al. 2020) changes sign, listed in Table 1.

Mean squared correlation: By analogy to $\langle\xi\rangle_{\theta_a}^{\theta_b}$ and $S_{1/2}$, we also evaluate the averge

$$\langle\xi^2\rangle_{\theta_a}^{\theta_b} = \frac{1}{\mu_a - \mu_b} \int_{\mu_b}^{\mu_a} \xi^2(\theta)\, d\mu \tag{6}$$

over the same annuli defined in Table 1. The quadratic statistic amplifies deviations that are coherent across multipoles. It cannot be cancelled by sign changes of $\xi(\theta)$, and it is thus ideally suited to quantify (lack of) correlation within the interval.

2.2. *Observational measurement*

The experimental values of each statistic are computed from the power spectrum[1] C_ℓ derived by the Commander component separation algorithm in the multipole range $2 \leq \ell < 30$ from the Planck 2018 temperature data (Aghanim et al. 2020). We have verified that including the higher order terms $\ell \geq 30$ derived from different implementations of the likelihood (Aghanim et al. 2020) has minimal effect on our results.

The antipodal correlation can be trivially derived from (4) by substituting $P_\ell(-1) = (-1)^\ell$.

For the interval-averaged correlation function, one may

[1] https://irsa.ipac.caltech.edu/data/Planck/release_3/ancillary-data/cosmoparams/COM_PowerSpect_CMB-TT-full_R3.01.txt

TABLE 2: Fiducial cosmological parameters (Aghanim et al. 2020).

Parameter	value
$\Omega_b h^2$	
$\Omega_c h^2$	
H_0	
τ	
n_s	
A_s	... FILL VALUES

integrate the Legendre polynomials analytically:

$$\langle \xi \rangle_{\theta_a}^{\theta_b} = \sum_\ell \frac{D_\ell}{2\ell(\ell+1)} \frac{I_\ell(\mu_a) - I_\ell(\mu_b)}{\mu_a - \mu_b}, \tag{7}$$

with

$$I_\ell(\mu) \equiv P_{\ell+1}(\mu) - P_{\ell-1}(\mu). \tag{8}$$

The mean squared correlation can be expressed a double sum

$$\langle \xi^2 \rangle_{\theta_a}^{\theta_b} = \frac{1}{\mu_a - \mu_b} \sum_{n,m} A_n A_m D_n D_m \, T_{nm}, \tag{9}$$

where

$$A_\ell \equiv \frac{2\ell+1}{2\ell(\ell+1)} \tag{10}$$

are normalisation constants, and

$$T_{nm} \equiv \sum_{r=0}^{\min(m,n)} \frac{A_r A_{m-r} A_{n-r} [J_{mnr}(\mu_a) - J_{mnr}(\mu_b)]}{A_{m+n-r}(2m+2n-2r+1)} \tag{11}$$

with

$$J_{mnr}(\mu) \equiv P_{m+n-2r+1}(\mu) - P_{m+n-2r-1}(\mu) \tag{12}$$

are given by the overlap integrals of the Legendre polynomials over the interval (Carlitz 1961)—a similar expression to the one obtained in (Copi et al. 2009), extended to different angles.

The classic $S_{1/2}$ statistic defined by Spergel et al. (2003) is equivalent to $S_{1/2} = \frac{3}{2}\langle \xi^2 \rangle_{60}^{180}$, and S_1 (Gruppuso 2014) corresponds to $S_1 = 2\langle \xi^2 \rangle_{0}^{180}$.

The values of all our statistical descriptors, derived from the observed C_ℓ, are quoted on the second column of Table 3.

2.3. *Prediction of the ΛCDM model*

As a fiducial example of the ΛCDM cosmology, we adopt the best-fit model in the publicly available Planck 2018 MCMC chain corresponding to the CamSpec high-multipole TT likelihood combined with the low-ℓ temperature and polarization likelihoods (`base_CamSpecHM_TT_lowl_lowE`, retrieved from the Planck Legacy Archive[2]). Its main parameters are summarised in Table 2.

The best-fit power spectrum $D_\ell^{\rm bf}$ are read from the `.minimum.theory_cl` file of the Planck chain, which contains the spectrum evaluated at the maximum-posterior parameter point reported in Table 2. From this input spectrum, $N = 1000$ independent Gaussian random realisations i with $\langle a_{\ell m}^{(i)} \rangle = 0$ and $\langle |a_{\ell m}^{(i)}|^2 \rangle = C_\ell^{\rm bf}$ are generated to account for cosmic variance. A minimal exploration of the ΛCDM parameter space based on different MCMC chains is briefly discussed in Appendix A.

The full analysis pipeline[3] is run for each random realisation. Following the methodology of Sullivan et al. (2024), we evaluate the theoretical $C_\ell^{(i)}$ spectrum from the random $a_{\ell m}^{(i)}$. Then, a full-sky temperature map is synthesized at $N_{\rm side} = 2048$ using `hp.synfast`, applying the appropriate beam transfer function b_ℓ and the HEALPix pixel window function p_ℓ. At $\ell_{\rm max} = 30$ the beam and pixel window corrections are numerically negligible, but they are included for consistency with the Planck data reduction. The pseudo-C_ℓ are recovered with `hp.anafast`, and the resulting spectrum is finally deconvolved by $b_\ell^2 p_\ell^2$ to obtain an estimate $\hat{D}_\ell^{(i)}$ in the same space as the published Planck values.

The predicted correlation function $\xi(\theta)^{(i)}$ and the derived statistics are evaluated analytically using Equations (4), (7), and (9). Median values over the N random realisations, as well as $16-84$ percentile intervals, are listed in Table 3.

2.4. *Statistical analysis*

For each statistic k, we compute the individual p-value p_k as the one-sided empirical quantile of the observed value within the simulated ensemble: $p_k = F_k$, where F_k is the fraction of realisations where the prediction is less than or equal to the observed value. The consistency between theory and data can be expressed in terms of Gaussian-like standard deviations as $n_\sigma = \Phi^{-1}(1 - p_{\rm global}/2)$, where Φ^{-1} is the probit function. Individual p-values and n_σ are reported for all our test statistics in the last column of Table 3.

To obtain a calibrated global significance, the values of ξ_{180}, as well as $\langle \xi \rangle_{\theta_a}^{\theta_b}$ and $\langle \xi^2 \rangle_{\theta_a}^{\theta_b}$ over the different intervals, must be combined together in a way that takes into account their correlation structure and the look-elsewhere effect. Since they are all derived from the same realisation of D_ℓ, their p-values are correlated by construction. Reporting only the most extreme individual p-value would overestimate the statistical significance of any potential tension, while treating them as independent and applying a simple Bonferroni correction would be overly conservative.

As a compromise, we apply the Aggregated Cauchy Association Test (ACAT; Liu & Xie 2020)

$$T = \sum_{k=1}^{11} w_k \tan\left[\left(\tfrac{1}{2} - p_k\right)\pi\right],$$

with equal weights $w_k = 1/13$, reflecting no a priori preference for any particular angular interval or statistic. Under the null hypothesis (that the observed value corresponds to a random realisation of the ΛCDM model), T follows a standard Cauchy distribution regardless of the correlation among the p_k, and the combined p-value

[2] https://pla.esac.esa.int/

[3] https://github.com/PhyAMR/CMB

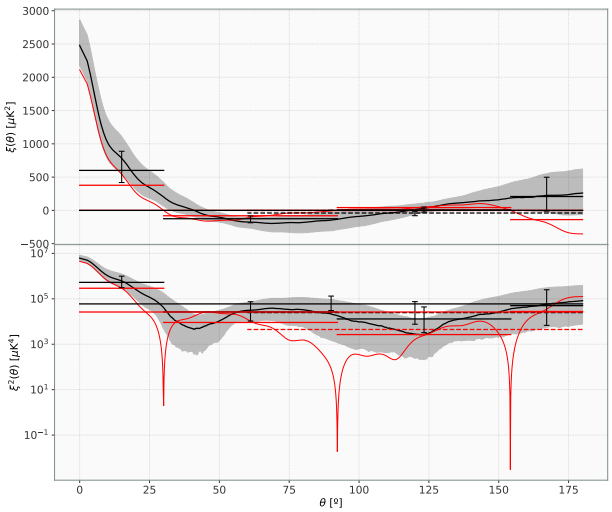


FIG. 1.—: Correlation function $\xi(\theta)$ (top) and its square $\xi^2(\theta)$ (bottom, log scale) for the fiducial model (Table 2), showing the median and 68% band of the theoretical ensemble (black line and grey band) together with the per-interval $\langle\xi\rangle_{\theta_a}^{\theta_b}$ (top) and $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ (bottom) diagnostics across the partitioned angular intervals of Table 1. Red segments mark the observed Planck value and black segments the model median in each interval, with vertical error bars showing the model's 68% confidence region; the classic $[60°, 180°]$ interval is displayed in dashed style.
... DOTTED FOR $0 - 180°$?
(REMOVE FROM TOP PANEL)
... LIMIT BOTTOM Y-AXIS TO $\xi^2 > 10$?
... AXIS LABELS FONT SIZE (in all figures)

may be obtained analytically as

$$p_{\rm global} = \frac{1}{2} - \frac{\arctan T}{\pi}.$$

The Cauchy combination is exact under arbitrary dependence of the individual p_k, including the strong correlations expected between statistics computed over overlapping angular ranges.

3. RESULTS

Figure 1 shows the adopted partition together with the observed and predicted correlation function $\xi(\theta)$ (top panel) and its square $\xi^2(\theta)$ (bottom panel). As explained in the previous section, our fixed intervals track the sign changes of the data rather than those of the ΛCDM prediction. Planck measurements are shown in red, while black lines display the theoretical predictions based on random realisations drawn from the the best-fit ΛCDM model (Table 2) to account for cosmic variance.

A more detailed view of the probability distribution of the different statistics predicted by our $N = 1000$ random realisations of the fiducial ΛCDM model is provided in Figures 2 and 3, compared to the corresponding values inferred from Planck data. Figure 2 focuses on the antipodal correlation ξ_{180}, whereas the different panels in Figure 3 display the results for $\langle\xi\rangle_{\theta_a}^{\theta_b}$ (left column) and $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ (right column) for each angular interval. A quantitative statistical summary (including the median, $16-84$ percentiles, and p-value of the observational measurement for each individual descriptor) is reported in Table 3.

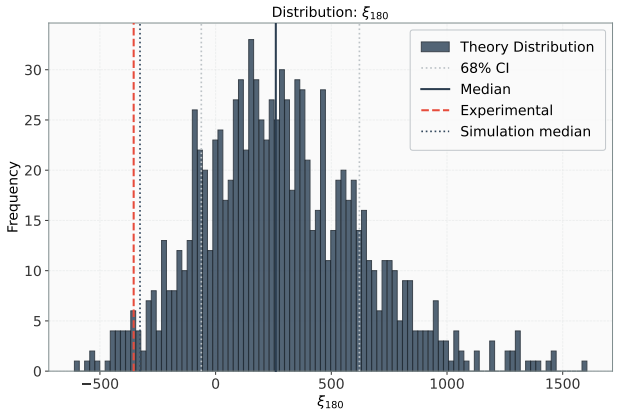


FIG. 2.—: Antipodal correlation ξ_{180} predicted by the fiducial ΛCDM cosmology (Table 2). The histogram illustrates the probability distribution over $N = 1000$ Monte Carlo realisation; the median and 68% confidence interval are shown by the black solid and dotted lines, respectively. Planck measurements are indicated by the red dashed line,
... WHAT IS THE THIRD DOTTED LINE?

The antipodal correlation ξ_{180} displays the largest discrepancy between theory and observations, with a p-value in the range between one and three per cent (about ~ 2 sigma). It is noteworthy that Planck data feature mild evidence for *anti-correlated* temperature fluctuations on opposite sides of the sky rather than purely independent, uncorrelated values. However, the statistical significance of this signal is weak. Our experiments show that the cosmic variance associated to this descriptor, estimated as half the 68% confidence interval, is of the same order as the measured value, i.e. $\Delta\xi_{180} \sim |\xi_{180}| \sim 350\ \mu\mathrm{K}^2$.

We do not find any significant anomaly in the interval-averaged correlation function $\langle\xi\rangle_{\theta_a}^{\theta_b}$. It is evident from the histograms in Figure 3 that the observed values of this descriptor are always close to the 68% confidence interval, sometimes higher sometimes lower than the median theoretical prediction. Once again, the largest angular scales $(154 - 180°)$ display the most significant deviation, both in absolute, relative, and qualitative terms (i.e. sign reversal). Interestingly, we also notice a deficit of observed power on the smallest scales $(0 - 30°)$, albeit at very low statistical significance (of the order of only one sigma).

Lack of power is most accurately traced by the $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ descriptor. As can be readily seen in Figures 1 and 3, as well as in Table 3, we find that all measured values are systematically below the median of the theoretical distribution. We would like to highlight that this is not a feature of the large angular scales. As a matter of fact, it is evident from Figure 1 that the strongest lack of correlation is observed at small $(0 - 30°)$ and intermediate $(60 - 120°)$ angles. At the largest scales, traced by the outermost interval, the experimental value of $\langle\xi^2\rangle_{154}^{180}$ is comparable to the median prediction of the ΛCDM cosmology. Our results are much better described as "anti-correlation" or "sign reversal" than "lack of power" or

TABLE 3: Experimental measurement of each descriptor from Planck data, median with 68% confidence interval (16 and 84 percentiles as subscript/superscript) of the ΛCDM model predictions, and one-sided p-value of the observed value. The global ACAT Cauchy statistic T and its p-vaule are provided in the model headers.

... UNITS
... SIGNIFICANT FIGURES
... REMOVE $\langle\xi\rangle_0^{180}$

"lack of correlation".

... Say something about S (1/2 and 1)?

3.1. *MCMC Analysis*

In the MCMC analysis the theoretical sampling distribution is constructed directly from the Planck posterior, propagating full parameter uncertainty through to the derived statistics. The 1000 posterior samples drawn from the CamSpec ΛCDM chain each yield a distinct D_ℓ spectrum, whose distribution reflects both cosmic variance and parameter degeneracies.

Figure 4 shows the resulting distributions of $\langle\xi\rangle_{\theta_a}^{\theta_b}$ (left column) and $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ (right column), and Figure 5 shows the analogous distribution for ξ_{180}. The broader width of the MCMC-derived histograms compared to the best-fit case reflects the additional spread due to cosmological parameter uncertainty.

We see the same trend as in the best-fit analysis, with the observed values of $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ consistently below the median of the distribution. The overall tension with ΛCDM is slightly increased in the MCMC analysis, mainly due to the additional contribution of parameter uncertainty, which broadens the theoretical distribution and shifts the median slightly.

The statistical summary, including the median, 68% credible interval, observed value, and empirical p-value for each statistic, is reported in Table 3.

4. DISCUSSION

... Compare new descriptors to classical S
... Benefits of ACAT
... Discrepancies are not significant
... Ω_k
... Dependence on angular scales
... Dependence on mask

The results for alternative likelihoods and models with free curvature Ω_k are reported in Appendix A. The dependence of the results on the likelihood choice (CamSpec vs. Plik) provides a useful cross-check, while the curvature-extended models test whether a departure from flatness could alleviate the observed tensions.

As noted by (Kim & Naselsky 2011), a lack of correlation is observed across all angular scales—including large, intermediate, and small angles. This is reflected in the fact that the observed values of $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ fall below the distribution's median for all intervals. This behavior is observed across all models and likelihoods examined, as detailed in Appendix A. While the significance of the individual statistics is not particularly high, the consistent pattern of low $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ values across all intervals suggests a coherent suppression of correlations that is not confined to a flat model, but rather extends to models with free curvature.

—

The statistics studied here are designed to probe complementary aspects of the large-angle correlation function: its integrated power, its mean level over specific angular ranges, and its value at antipodal separation. The choice is guided by the existing literature on low-ℓ anomalies, though as noted by Efstathiou et al. (2010), some statistics, including $S_{1/2}$, were identified a posteriori from the data, which inflates their apparent significance. Studying multiple statistics simultaneously, and combining their p-values through a global test, is therefore essential to obtain a reliable assessment of potential tensions with the ΛCDM scenario.

We note that defining the intervals from a feature of the data under test is itself a form of a posteriori choice, of the kind discussed by Efstathiou et al. (2010) (Section 1).

...

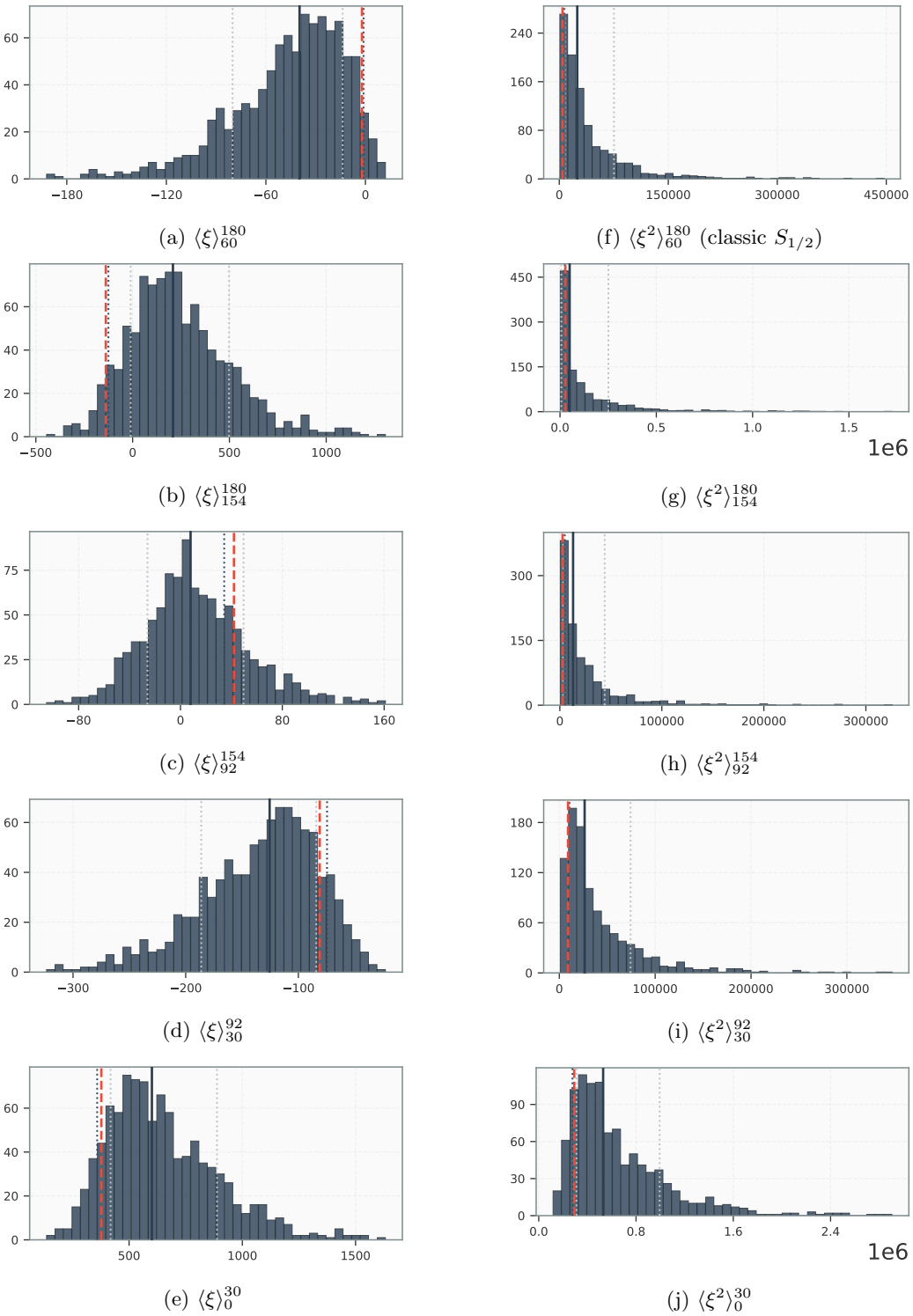
Rather than partitioning $[0°, 180°]$ into fixed, evenly spaced angular bins, the boundaries were placed at the zeros of the correlation function measured from the observed Planck multipoles. A fixed, arbitrary binning would not be representative of the measured $\xi(\theta)$: it could mix angular ranges in which the data is predominantly positive with ranges in which it is predominantly negative, and the resulting cancellation within $\langle\xi\rangle_{\theta_a}^{\theta_b}$ would dilute any genuine mismatch between the observed and predicted correlation function. Partitioning at the observed zero-crossings instead guarantees that each interval corresponds to a single coherent sign region of the measured signal, so that $\langle\xi\rangle_{\theta_a}^{\theta_b}$ directly compares the predicted and observed mean level lobe by lobe, and $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ probes the predicted and observed power within the same regions.

...

This affects only the two interval-based statistics, $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ and $\langle\xi\rangle_{\theta_a}^{\theta_b}$; the antipodal statistic ξ_{180}, which yields the strongest individual tension found in this work (Table 3), does not depend on this choice at all. The theoretical sampling distributions of Section **??** are evaluated on exactly the same intervals, so the reported p-values remain valid tests of whether the observed value is unusual within each interval; what they do not capture is the additional, harder-to-calibrate look-elsewhere effect of having selected the interval boundaries from the same dataset being tested. We mitigate this only partly, by retaining the classic $[60°, 180°]$ interval for direct comparison with the literature and by combining all eleven statistics through ACAT rather than reporting any single interval-based result in isolation.

5. CONCLUSIONS

We have studied the large-angle CMB angular correlation function and its derived statistics using Planck 2018 data and two complementary ensembles for the theoretical sampling distributions. The statistics examined, the quadratic $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$, the linear $\langle\xi\rangle_{\theta_a}^{\theta_b}$, and the antipodal



(a) $\langle \xi \rangle_{60}^{180}$ (f) $\langle \xi^2 \rangle_{60}^{180}$ (classic $S_{1/2}$)

(b) $\langle \xi \rangle_{154}^{180}$ (g) $\langle \xi^2 \rangle_{154}^{180}$

(c) $\langle \xi \rangle_{92}^{154}$ (h) $\langle \xi^2 \rangle_{92}^{154}$

(d) $\langle \xi \rangle_{30}^{92}$ (i) $\langle \xi^2 \rangle_{30}^{92}$

(e) $\langle \xi \rangle_{0}^{30}$ (j) $\langle \xi^2 \rangle_{0}^{30}$

FIG. 3.—: Interval-averaged correlation $\langle \xi \rangle_{\theta_a}^{\theta_b}$ (left column) and integrated squared correlation $\langle \xi^2 \rangle_{\theta_a}^{\theta_b}$ (right column) for each angular interval. Line styles in every panel are identical to Figure 2.

ξ_{180}, probe distinct aspects of the large-angle temperature field.

Both ensembles provide complementary perspectives on the significance of the large-angle anomalies. The best-fit ensemble isolates the contribution of cosmic variance, while the MCMC ensemble captures the full parameter uncertainty and the cosmic variance.

The large-angle suppression of the correlation function, most evident in ξ_{180}, is reproducible across both ensembles and across all likelihood and curvature variants tested (Appendix A), but it is not statistically significant once the eleven correlated statistics are combined through the ACAT global test: the resulting tension with ΛCDM is $\sim 1.1\sigma$ for the primary `base_CamSpecHM_TT_lowl_lowE` model (Table 3).

We conclude that current Planck data do not provide significant evidence against the standard cosmological model from the large-angle correlation function.

REFERENCES

Aghanim, N., Akrami, Y., Ashdown, M., et al. 2020, Astronomy amp; Astrophysics, 641, A5, doi: 10.1051/0004-6361/201936386

Carlitz, L. 1961, Archiv der Mathematik, 12, 334, doi: 10.1007/BF01650571

Chiocchetta, C., Gruppuso, A., Lattanzi, M., Natoli, P., & Pagano, L. 2020, Journal of Cosmology and Astroparticle Physics, 2021, doi: 10.1088/1475-7516/2021/08/015

Copi, C., Huterer, D., Schwarz, D., & Starkman, G. 2006, Physical Review D, 75, 023507, doi: 10.1103/physrevd.75.023507

—. 2013, Monthly Notices of the Royal Astronomical Society, 451, 2978, doi: 10.1093/mnras/stv1143

Copi, C. J., Huterer, D., Schwarz, D. J., & Starkman, G. D. 2009, Monthly Notices of the Royal Astronomical Society, 399, 295–303, doi: 10.1111/j.1365-2966.2009.15270.x

Dipanshu, Souradeep, T., & Hirve, S. 2023, The Astrophysical Journal, 954, doi: 10.3847/1538-4357/ace895

Efstathiou, G., Ma, Y.-Z., & Hanson, D. 2010, Monthly Notices of the Royal Astronomical Society, 407, 2530, doi: 10.1111/j.1365-2966.2010.17081.x

Froes, A. L. D., Pereira, T. S., Bernui, A., & Starkman, G. 2015, Physical Review D, 92, doi: 10.1103/physrevd.92.043508

Gruppuso, A. 2014, Monthly Notices of the Royal Astronomical Society, 437, 2076, doi: 10.1093/mnras/stt1937

Hinshaw, G., Banday, A. J., Bennett, C. L., et al. 1996, The Astrophysical Journal, 464, L25–L28, doi: 10.1086/310076

Joshi, N., Jhingan, S., Souradeep, T., & Hajian, A. 2009, Physical Review D, 81, 083012, doi: 10.1103/physrevd.81.083012

Kim, J., & Naselsky, P. 2011, The Astrophysical Journal, 739, 79, doi: 10.1088/0004-637X/739/2/79

Lewis, A. 2000, CAMB: Code for Anisotropies in the Microwave Background, https://github.com/cmbant/CAMB

Liu, Y., & Xie, J. 2020, Journal of the American Statistical Association, 115, 393, doi: 10.1080/01621459.2018.1554485

Pontzen, A., & Peiris, H. V. 2010, Physical Review D, 81, doi: 10.1103/physrevd.81.103008

Sarkar, D., Huterer, D., Copi, C., Starkman, G., & Schwarz, D. 2010, Astroparticle Physics, 34, 591, doi: 10.1016/j.astropartphys.2010.12.009

Spergel, D. N., Verde, L., Peiris, H. V., et al. 2003, The Astrophysical Journal Supplement Series, 148, 175, doi: 10.1086/377226

Sullivan, R. M., Hergt, L. T., & Scott, D. 2024, Methods for CMB map analysis. https://arxiv.org/abs/2410.12951

Zhang, S. 2012, The Astrophysical Journal, 748, L20, doi: 10.1088/2041-8205/748/2/120

APPENDIX

A. ADDITIONAL MODELS: STATISTICAL TABLES

...

The spread of the resulting ensemble is driven entirely by cosmic variance at fixed cosmological parameters, with no contribution from parameter uncertainty. The width of its histograms is therefore a lower bound on the total variance: the additional width seen in the MCMC histograms is attributable purely to parameter uncertainty, and comparing the two ensembles provides a clean decomposition of the two contributions.

This provides posterior samples of the six standard ΛCDM parameters $\{\Omega_b h^2,\ \Omega_c h^2,\ H_0,\ \tau,\ n_s,\ A_s\}$ along with a suite of derived quantities. The Plik-based TT likelihood and two curvature-extended runs are considered in .

For each posterior sample the theoretical power spectrum D_ℓ is computed with CAMB (Lewis 2000).

Two independent strategies are used to construct the theoretical sampling distribution. The full analysis pipeline is publicly available here. The exact numerical results reported in this paper can be reproduced from the uploaded output files; because the pipeline uses a random seed for each realisation, re-running it from scratch will reproduce the same statistical conclusions but not bit-identical histograms.

1. **MCMC ensemble.** For each of the 1000 samples drawn from the tail of the Planck posterior chain, specifically the last 1000 rows after GetDist's default burn-in removal, the cosmological parameters are passed to CAMB to compute a theoretical spectrum $D_\ell^{\rm th}$. Each spectrum is then passed through the simulation pipeline described below, producing one realisation of cosmic variance on top of the parameter-determined $D_\ell^{\rm th}$. The resulting ensemble therefore captures both parameter uncertainty and cosmic variance simultaneously, constituting a Bayesian predictive distribution for each statistic.

—

The tables in this appendix report the full statistical summary for three additional Planck models, each in a single table with the MCMC and best-fit ensembles side by side (as in Table 3). All quantities are defined as in the main text. Model names follow the Planck convention: `omegak` denotes a run with free spatial curvature; `plikHM` denotes the Plik high-multipole likelihood; `TTTEEE` denotes a joint temperature and polarization fit.

TABLE 4: Statistical summary for `base_plikHM_TT_lowl_lowE`.

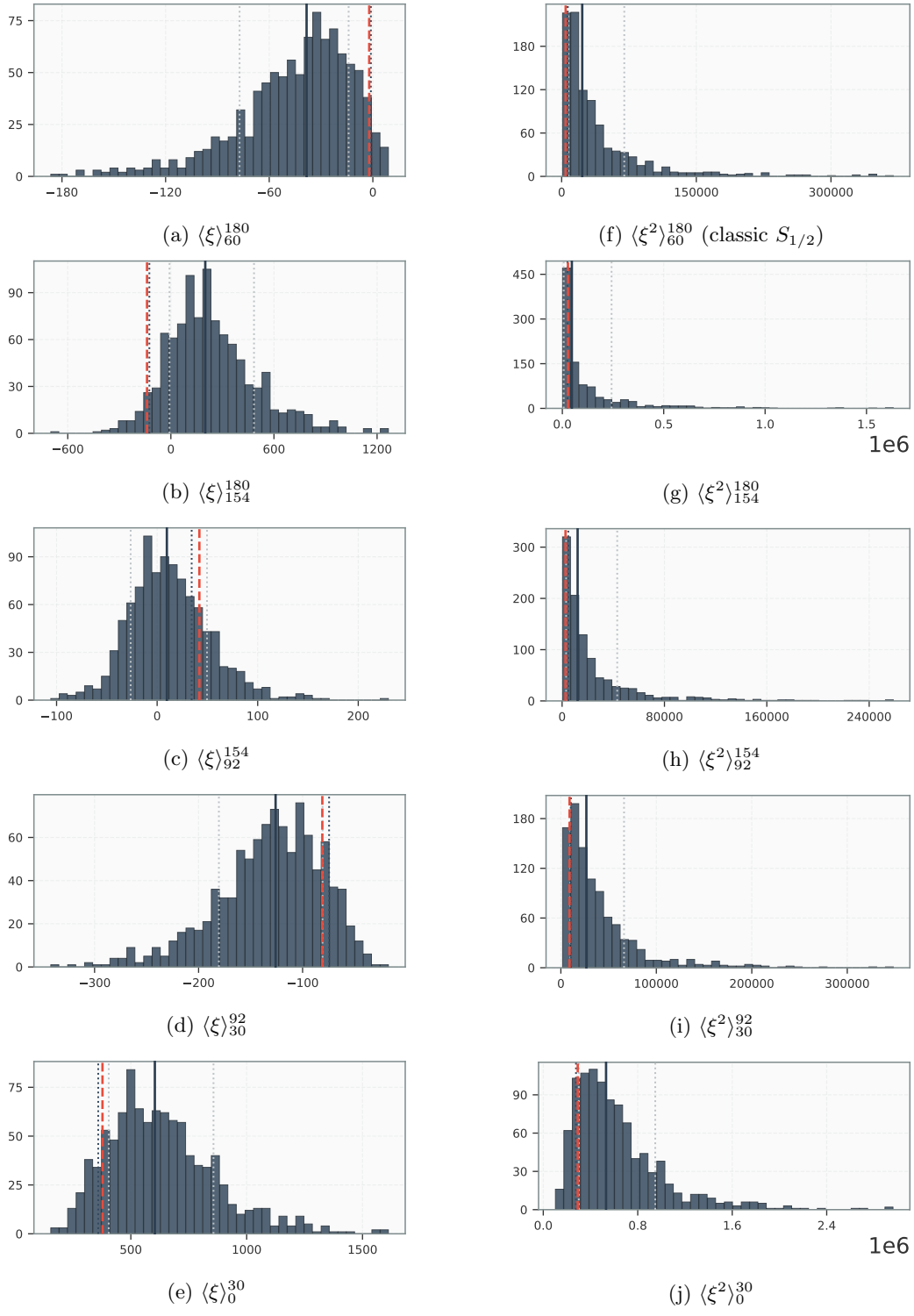


FIG. 4.—: MCMC posterior distributions for model `base_CamSpecHM_TT_lowl_lowE`: the interval-averaged correlation $\langle\xi\rangle_{\theta_a}^{\theta_b}$ (left column) and the integrated squared correlation $\langle\xi^2\rangle_{\theta_a}^{\theta_b}$ (right column), for each of the five angular intervals of Table 1, each panel showing the distribution over 1000 posterior samples from the Planck CamSpec chain. Line styles are as in Figure 3.

TABLE 5: Statistical summary for `base_omegak_CamSpecHM_TT_lowl_lowE`.

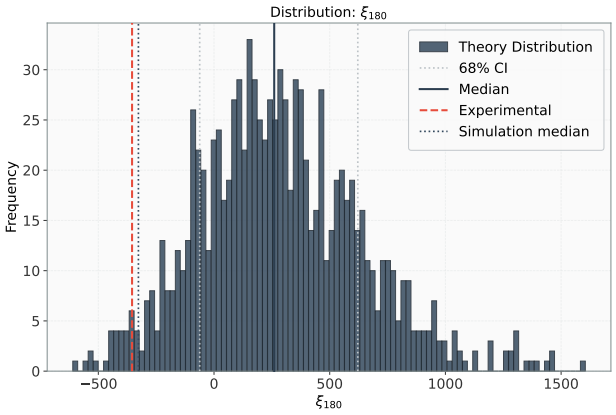


FIG. 5.—: MCMC posterior distribution of ξ_{180} for model `base_CamSpecHM_TT_lowl_lowE`.

TABLE 6: Statistical summary for `base_omegak_plikHM_TTTEEE_lowl_lowE`.

This paper was built using the Open Journal of Astrophysics LaTeX template. The OJA is a journal which provides fast and easy peer review for new papers in the `astro-ph` section of the arXiv, making the reviewing process simpler for authors and referees alike. Learn more at http://astro.theoj.org.